Finite terms: complete simplification, running coupling and DGLAP extracted, final result

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1 Summary

Input. `Finite_terms_1.tex` is identical, byte for byte, to the `Finite_terms.tex` of `Finite_terms_exact.pdf`. This note replaces that one. As there:

- $a \equiv \frac{k^+}{\vee}$ is kept exactly;
- the only limit taken is $\Lambda \to 0$, wherever it exists;
- every expression is checked numerically against your file.

What is new.

1. **Consistent with the corrected evolution terms.** The evolution is now $\mathcal{N}\,\frac{1}{2}\log\frac{\vee}{\Lambda}(\mathcal{J}+\mathcal{B})$ (Secs. 3–5 of `Evolution_terms_corrected.tex`).
   - Every row is split exactly as $C_{\rm row}\log\frac{\vee}{k^+} + R_{\rm row}(a)$, with $R_{\rm row}$ finite for $a \to 0$ (Sec. 6).
   - The coefficients add up to $\sum C_{\rm row} = \mathcal{B} - \mathcal{J}$. So in every row $\log\frac{\vee}{k^+}$ in front of $C_{\rm row}$ is replaced by

$$L_* \equiv \log\frac{\sqrt{\vee\Lambda}}{k^+} = \log\frac{\vee}{k^+} - \frac{1}{2}\log\frac{\vee}{\Lambda}.$$

     This is finite and is zero at mid rapidity.

- Summed over the rows, these terms are the remainder $\log\frac{k^+}{\sqrt{\vee\Lambda}}(\mathcal{J}-\mathcal{B})$ of Sec. 5 of the corrected evolution file (counted once).

2. **c.c. checked**, as you say: Group II rows are self-conjugate (no c.c.); Group I and Group III rows are not, and get +c.c. (Sec. 2.1).

3. **Rows with the same kernel combined.** Their colour factors collapse:
   - Group I Row IV, Group III Rows I, II and Group II Row I part 1 share the kernel $\mathcal{K}_{\rm IV}$;
   - Group III Rows III(1), V, VI share the kernel $\mathcal{K}_C$; their colour becomes one product of three dipole-like factors (Sec. 8).

4. **Three ρ's reduced completely.** Every three-ρ row is written as its fully symmetric three-ρ part (compact) plus its two-ρ part. The two-ρ part is written out, merge pattern by merge pattern (Sec. 9).

5. Group II Row I part 2 is collected into one bracket (coefficients of λ, λ', L_*, $\log(1-a)$, 1).

Results.

- **Running coupling** (Group I Row II part 1 + Row V 2b UV, both with c.c.): the β_0 terms cancel exactly. Left over (Eq. (4.3)):

$$\frac{\alpha_s}{2\pi}N_c\Big[\tfrac{67}{9}-\tfrac{\pi^2}{3}\Big]\mathrm{LO}\ \text{(CMW constant)},\qquad -\frac{\alpha_s}{2\pi}N_c\log\Big(1-\frac{k^+}{\vee}\Big)\log\Big(\frac{\vee}{k^+}-1\Big)\mathrm{LO}\ \text{(not running coupling)}.$$

  The second term is $-\log^2(1-a)-\log\frac{\vee}{k^+}\log(1-a)$ written as one product. It comes from the limit $\xi\geq\frac{k^+}{\vee}$ of the dressed one-ρ state, like the pole of Q1, and vanishes for $k^+\ll\vee$ (`Finite_terms_singlet.pdf`, Sec. 1).

- **DGLAP** (Group II Row IV, no c.c.): the pole is absorbed in $D_{h/g}$ with $\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}P_{gg}(\xi)$. The finite part is Eq. (5.5); the linear-polarization term is Eq. (5.6).

- **Final result**: Eq. (11.1).

Still open (unchanged from `Finite_terms_exact.pdf`; your decision is needed, Sec. 12):

- **Q1:** the pole $\frac{\alpha_s}{2\pi}\frac{2}{\epsilon}N_c\log(1-\frac{k^+}{\vee})$LO is left over;
- **Q2:** the four-ρ finite term.

2 Notation and conventions

- $a\equiv\frac{k^+}{\vee}\in(0,1)$, $L_\vee\equiv\log\frac{\vee}{k^+}=-\log a$, $L_*\equiv\log\frac{\sqrt{\vee\Lambda}}{k^+}=L_\vee-\frac12\log\frac{\vee}{\Lambda}$.
- $\ell_r\equiv\log\frac{r^2\mu^2_{\overline{MS}}}{c_0^2}=2\gamma_E+\log\frac{r^2\mu^2_{\overline{MS}}}{4}$, $c_0=2e^{-\gamma_E}$, $\hat{E}_r\equiv\frac{2}{\epsilon}+\ell_r$.
- $A\equiv(x-z)^2$, $B\equiv(x-w)^2$, $A'\equiv(x'-z)^2$, $B'\equiv(x'-w')^2$, $D\equiv B'A-BA'$.
- $\mathcal{K}(v)\equiv\frac{v}{v^2}$; $X'\equiv\mathcal{K}(x'-w')$, $Y\equiv\mathcal{K}(y-z)$, $X_w\equiv\mathcal{K}(x-w)$, $X_z\equiv\mathcal{K}(x-z)$, $Z_w\equiv\mathcal{K}(z-w)$.
- $\lambda\equiv\log\big[(1-a)\frac{A}{B}+a\big]$, $\lambda'\equiv\log\big[(1-a)\frac{A'}{B'}+a\big]$, $\lambda_2\equiv\log\big|\frac{A}{B}-a\big|$. All three are finite for $a\to0$.
- The brackets of your Rows IV (Group I), I (Groups II, III):

$$\mathbb{T}^{ij}\equiv\frac{\delta^{ij}}{2(z-w)^2}+\frac{(x-z)^j(z-w)^i}{(z-w)^2(x-z)^2}-\frac{(x-z)^j(x-w)^i}{(x-w)^2(x-z)^2}-\frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2},$$
$$\mathbb{L}^{ij}\equiv\frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2}+\frac{(x-z)^j(x-w)^i}{2(x-z)^2(x-w)^2}.$$

- Colour: $\mathcal{C}_L^a\equiv\big[U^{ab'}(x')-U^{ab'}(w')\big]\rho^{b'}(x')$, $\mathcal{R}^{ab}(w,z,x)\equiv f^{cbd}U^{ac}(w)\rho^d(x)-f^{acd}U^{bc}(z)U^{de}(x)\rho^e(x)$.
- LO, with $\mathcal{O}\equiv[U^{e'c}(y')-U^{e'c}(x')][U^{ec}(y)-U^{ce}(x)]\rho^{e'}(x')\rho^e(x)$:

$$\mathrm{LO}\big[f\big]\equiv\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2k^+}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)}\,f\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\,\mathcal{O},\qquad\mathrm{LO}\equiv\mathrm{LO}[1],$$
$$\mathrm{LO}_\xi\big[f\big]\equiv\text{the same with }e^{-ik\cdot(y'-y)}\to e^{-ik\cdot(y'-y)\frac{1}{\xi}}.$$

  In LO$[\ell_X]$, $X=x-y$ (the emission distance); $\Delta\equiv y'-y$.

- $\mathcal{N} \equiv \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}$; $p_8 \equiv \frac{ig^4}{8\pi^5 k^+}$, $p_{16} \equiv \frac{ig^4}{16\pi^5 k^+}$.
- $\mathcal{J}$, $\mathcal{B}$: the brackets of Secs. 3 and 4 of `Evolution_terms_corrected.tex` (JIMWLK⊗LO and BFKL⊗LO, units of $\mathcal{N}$).
- $E_1(z) = \int_z^\infty \frac{e^{-t}}{t}dt$ (principal branch); $\mathrm{Ein}(z) = \int_0^z \frac{1-e^{-t}}{t}dt = E_1(z) + \log z + \gamma_E$ (a-independent).
- $U^{bc} = U^{cb}$, as in your files.

2.1 The c.c.

For a Group I or Group III term T, written with the phase $e^{-ik\cdot(w'-w)}$ (or $e^{-ik\cdot(y'-w)}$):

$$T + \text{c.c.}, \qquad \text{c.c.} \equiv \Big[T\Big]^*_{w\leftrightarrow w'} \quad (\text{resp. } w \leftrightarrow y').$$

- * means $i \to -i$ in the prefactors, $E_1(i\beta) \to E_1(-i\beta)$, $\log(i\kappa) \to \log(-i\kappa)$, and the order of the ρ's reversed.
- The exchange of the two points of the measured gluon restores the phase.

Numerically, at random configurations (classical charges), with $\tilde{c}$ = the configuration with the amplitude and conjugate points exchanged:

$\lvert T(c) - T(\tilde{c})^*\rvert/\lvert T(c)\rvert$	
G2-I(1) $[w \leftrightarrow w']$	0 (self-conjugate)
G2-I(2) $[w \leftrightarrow w', x \leftrightarrow x']$	1.7×10^{-15} (self-conjugate)
G1-IV $[w \leftrightarrow w', x \leftrightarrow x']$	1.9×10^{0} (not self-conjugate)
G1-IV $[w \leftrightarrow w']$	1.0×10^{0} (not self-conjugate)
G3-I $[w \leftrightarrow w', x \leftrightarrow x']$	1.9×10^{0} (not self-conjugate)
G3-I $[w \leftrightarrow w']$	1.0×10^{0} (not self-conjugate)
G2-IV LO structure $[y \leftrightarrow y',\ x \leftrightarrow x']$	0 (self-conjugate)

So Group II needs no c.c., as you say: Row I part 1 contains both orders (blocks A and B, B = c.c. of A), Row I part 2 is symmetric under $(x,w) \leftrightarrow (x',w')$, Row IV under $(x,y) \leftrightarrow (x',y')$. Group I and Group III rows are not self-conjugate; their c.c. is added everywhere below.

3 Step 1: $\Lambda \to 0$

Exactly in a:

$$\log\frac{(\vee - k^+)A + k^+B}{\Lambda A + k^+B} \to L_\vee + \lambda, \qquad \log\Big|\frac{\vee A - k^+B}{(\Lambda + k^+)A - k^+B}\Big| \to L_\vee + \log\Big|\frac{A - aB}{A - B}\Big|,$$

$$\log\Big|\frac{k^+B - \vee A}{k^+B - \Lambda A}\Big| \to L_\vee + \lambda_2, \qquad \log\Big|\frac{(k^+ - \Lambda)A - k^+B}{\Lambda A - k^+B}\Big| \to \log\Big|\frac{A - B}{B}\Big|,$$

$$\log\frac{[k^+(x-w) - (k^+ - \Lambda)(x-z)]^2}{[k^+(x-w) - \Lambda(x-z)]^2} \to \log\frac{(z-w)^2}{B}, \qquad \log\frac{\vee}{k^+ \pm \Lambda},\ -\log\frac{k^+ - \Lambda}{\vee} \to L_\vee,$$

$$\frac{1}{\vee} - \frac{1}{k^+ + \Lambda} \to -\frac{1-a}{k^+}, \qquad \log\Big(1 - \frac{\Lambda}{k^+}\Big) \to 0, \qquad \mathcal{A} \to \mathcal{A}_0,$$

with $\mathcal{A}_0 = \frac{1}{r}\big[\arctan\frac{A-(x-w)\cdot(x-z)}{r} + \arctan\frac{(x-w)\cdot(x-z)}{r}\big]$, $r = \sqrt{AB - [(x-w)\cdot(x-z)]^2}$.

- The logarithms of Group I Row III are principal values ($\log|\dots|$).
- The $\frac{1}{\Lambda}$ terms of Group I Row III cancel.
- Plus prescriptions start at 0; $\frac{k^+}{k^+ + \Lambda} \to 1$.

4 Step 2: running coupling

In units of LO (inside the integrals), the term and its c.c. ($X = x - y$, $X' = x' - y'$; Group I writes w, w' for y, y'; in 2b UV rename $y \to x$, $z \to y$):

$$\text{G1 Row II, part 1 + c.c.} = \frac{\alpha_s N_c}{4\pi}\,\mathrm{LO}\Big[\big(\hat{E}_X + \hat{E}_{X'}\big)\Big(\frac{11}{3} + 2L_\vee + \log\big(1-a\big)\Big)\Big] + \frac{\alpha_s N_c}{2\pi}\Big[\frac{67}{9} - \frac{\pi^2}{3} - \log^2(1-a) - L_\vee\log\big(1-a\big)\Big]\mathrm{LO}, \tag{4.1}$$

$$\text{G1 Row V, 2b UV + c.c.} = \frac{\alpha_s N_c}{2\pi}\,\text{LO}\Big[(\hat{E}_X+\hat{E}_{X'})\Big(-2L_\vee-\frac{11}{6}\Big)\Big]. \tag{4.2}$$

Your bracket $L_\vee+\log(\frac{\vee}{k^+}-1)$ is exactly $2L_\vee+\log(1-a)$.

- β_0**:** $\frac{11}{3}$ against $2\times\frac{11}{6}$, with pole, γ_E, μ and $\log X^2$, cancels exactly for every a (`sympy`: 0).
- **Left over**:

$$\boxed{\left.\frac{d^3N}{d^3k}\right|_{\rm RC} = \frac{\alpha_s}{2\pi}\,N_c\Big[\frac{67}{9}-\frac{\pi^2}{3}\Big]\text{LO}}, \qquad \boxed{\left.\frac{d^3N}{d^3k}\right|_{V_\vee} = -\frac{\alpha_s}{2\pi}\,N_c\,\log\Big(1-\frac{k^+}{\vee}\Big)\log\Big(\frac{\vee}{k^+}-1\Big)\text{LO}} \tag{4.3}$$

  - $-\log^2(1-a)-L_\vee\log(1-a) = -\log(1-a)\log\frac{1-a}{a}$ exactly. $V_\vee$ is not a running-coupling term: it is the part of Row II part 1 that comes from the limit $\xi\geq\frac{k^+}{\vee}$ of the dressed one-ρ state, the same limit that gives the pole of Q1 (`Finite_terms_singlet.pdf`, Sec. 1). It is $\approx a\log\frac{1}{a}$ for $a\to 0$.
  - $L_\vee\log(1-a)$ is not a large logarithm: it vanishes for $a\to 0$ and for $a\to 1$. So it stays as it is (Sec. 6).
  - The terms $(\hat{E}_X+\hat{E}_{X'})(2L_\vee+\log(1-a))$ and $-4L_\vee(\hat{E}_X+\hat{E}_{X'})$ are not running coupling. They combine with Group II Row IV (Sec. 5).
- **Scale of α_s in LO:** $\alpha_s=\alpha_s(\mu_F)$. The LO coupling counterterm is the $\frac{\beta_0}{2}\delta(1-\xi)$ part of the fragmentation counterterm.

5 Step 3: DGLAP and the poles

Group II Row IV (no c.c.), lines 1–2. With $d_\perp=2-\epsilon$ your weight is $W=-\hat{E}_\Delta+O(\epsilon)$. Your subtraction at $\xi=1$ is exactly the standard plus distribution restricted to $\xi>a$:

$$\text{lines 1, 2} = \frac{\alpha_s}{2\pi}\int_a^1\frac{d\xi}{\xi^2}\,P^{(r)}_{gg,+}(\xi)\,\text{LO}_\xi\big[W\big]-\frac{\alpha_s}{2\pi}\,2N_cL_\vee\,\text{LO}\big[W\big], \qquad P^{(r)}_{gg,+}=2N_c\Big[\frac{\xi}{(1-\xi)_+}+\frac{1-\xi}{\xi}+\xi(1-\xi)\Big], \tag{5.1}$$

with $\int_a^1 d\xi\frac{f(\xi)}{(1-\xi)_+}=\int_a^1 d\xi\frac{f(\xi)-f(1)}{1-\xi}+f(1)\log(1-a)$. Your second line ($\log(1-\frac{k^+}{\vee})$) is exactly the last term.

Poles. In units of $\frac{\alpha_s}{2\pi}(-\frac{2}{\epsilon})N_c\,\text{LO}$ (the $\delta(1-\xi)$ terms):

source	coefficient
G2 Row IV, Eq. (5.1)	$-2L_\vee$
G1 Row II part 1 + c.c., Eq. (4.1)	$-\frac{11}{3}-2L_\vee-\log(1-a)$
G1 Row V 2b UV + c.c., Eq. (4.2)	$+\frac{11}{3}+4L_\vee$
charge renormalization of LO	$+\frac{11}{6}$
sum	$\frac{11}{6}-\log(1-a)$ (`sympy`: $11/6-\log(1-a)$)

$$\left.\frac{d^3N_g}{d^3k}\right|_{\rm pole} = \frac{\alpha_s}{2\pi}\Big(-\frac{2}{\epsilon}\Big)\Big[\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\,P_{gg}(\xi)\,\text{LO}_\xi\;-\;N_c\log\Big(1-\frac{k^+}{\vee}\Big)\text{LO}\Big], \qquad P_{gg}=P^{(r)}_{gg,+}+\frac{11N_c}{6}\,\delta(1-\xi). \tag{5.2}$$

- The first term is absorbed in $D_{h/g}$, exactly. The hadron convolution has the lower limit $\frac{k^+}{\vee}$.
- The second term is not absorbed (Q1):

$$\left.\frac{d^3N}{d^3k}\right|_{\rm rem} = \frac{\alpha_sN_c}{4\pi}\,\log\Big(1-\frac{k^+}{\vee}\Big)\,\text{LO}\big[\hat{E}_X+\hat{E}_{X'}\big]. \tag{5.3}$$

The finite parts. (`sympy`, difference 0.) The $L_\vee$ terms of Eqs. (4.1), (4.2), (5.1) add up to a pole-free term (coefficient of $L_\vee$: $\frac{\alpha_sN_c}{2\pi}(2\ell_\Delta-\ell_X-\ell_{X'})$):

$$\frac{\alpha_sN_c}{2\pi}\,\log\frac{\vee}{k^+}\,\text{LO}\Big[\log\frac{(y-y')^4}{(x-y)^2(x'-y')^2}\Big] \quad\longrightarrow\quad \boxed{\frac{\alpha_sN_c}{2\pi}\,L_*\,\text{LO}\Big[\log\frac{(y-y')^4}{(x-y)^2(x'-y')^2}\Big]} \tag{5.4}$$

(the replacement $L_\vee\to L_*$ is that of Sec. 6). DGLAP:

$$\boxed{\left.\frac{d^3N}{d^3k}\right|_{\rm DGLAP} = -\frac{\alpha_s}{2\pi}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\,P^{(r)}_{gg,+}(\xi)\,\text{LO}_\xi\big[\ell_\Delta(\mu_F)\big], \qquad \ell_\Delta=\log\frac{(y'-y)^2\mu_F^2}{c_0^2}} \tag{5.5}$$

- $\mu_F^2\frac{\partial}{\partial\mu_F^2}D_{h/g}=\frac{\alpha_s}{2\pi}P_{gg}\otimes D_{h/g}$.
- With $d_\perp=2$: $\ell_\Delta\to\ell_\Delta+1$.

Line 3 (linear polarization). $\Lambda \to 0$ only in the upper limit:

$$\frac{d^3N_{\rm lin}}{d^3k} = \frac{\alpha_s}{2\pi}\int_{k^+/\vee}^1 \frac{d\xi}{\xi^2}\, 2N_c\,\xi(1-\xi)\,\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2 k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\Big[X\cdot X' - 2(X\cdot\hat\Delta)(X'\cdot\hat\Delta)\Big]\mathcal{O}, \tag{5.6}$$

with $X = \mathcal{K}(x-y)$, $X' = \mathcal{K}(x'-y')$, $\hat\Delta = \frac{y'-y}{|y'-y|}$.

6 Step 4: the large logarithm $\log\frac{\vee}{k^+}$ and the corrected evolution

Split. Every row is, exactly,

$$\text{row} = C_{\rm row}\, L_\vee + R_{\rm row}(a), \qquad R_{\rm row}(a) \text{ finite for } a \to 0. \tag{6.1}$$

$C_{\rm row}$ is read off from the closed forms of Sec. 7, using

- $\log\frac{i\kappa}{a} = \log(i\kappa) + L_\vee$;
- $E_1(\frac{i\kappa}{a}) \to 0$;
- $\lambda \to \log\frac{A}{B}$, $\lambda_2 \to \log\frac{A}{B}$, $\log(1-a) \to 0$.

Test: $R_{\rm row}(a) = \text{row} - C_{\rm row}L_\vee$ at $a = 10^{-4}, 10^{-6}, 10^{-8}$ stays constant (relative spread $\le 8.2\times10^{-4}$, i.e. $O(a)$). A wrong $C_{\rm row}$ would make it grow like $L_\vee$.

row	$C_{\rm row}$ (with the row's prefactor, phase, colour, c.c.)
G1 Row II part 1, G1 Row V 2b UV, G2 Row IV	together: $\frac{\alpha_s N_c}{2\pi}\mathrm{LO}\big[\log\frac{(y-y')^4}{(x-y)^2(x'-y')^2}\big]$ (Sec. 5)
G1 Row II 2a	the whole of 2a (it is $\propto L_\vee$)
G1 Row III	$p_8\big[(X'\cdot X_w)(Y\cdot Z_w - Y\cdot X_z) + \frac{X'\cdot Y}{(z-w)^2} - \Sigma_4\big]$
G1 IV, G2 I(1), G3 I, II	$\mathcal{K}^{(C)}_{\rm IV} \equiv X'^i Y^j \mathbb{T}^{ij} - \frac{X'\cdot Y}{2(z-w)^2}$
G1 V(1)	$\kappa_b\, e^{-ik\cdot(w'-z)} + \kappa_q\, e^{-ik\cdot(w'-x)}$
G1 V(2a), V(2b fin)	$e^{-ik\cdot(w'-z)}\frac{s_3}{Y_z} + e^{-ik\cdot(w'-x)}\frac{s_6}{Y_x}$ (2b fin: $+2e^{-ik\cdot(w'-z)}\frac{X'\cdot Y\,(y-x)\cdot(y-z)}{(y-x)^2(x-z)^2}$)
G2 I(2)	$-\frac{[\dots]_{2b}}{BB'} + \frac{[\dots]_{2c}}{AA'}$
G3 III(1), V, VI	t_2
G3 III(2)	$-\frac{s}{k^+B}$

Sum. $\sum_{\rm rows} C_{\rm row} = \mathcal{N}(\mathcal{B}-\mathcal{J})$, after the three-$\rho$ symmetrization and the z integral of the pole terms. This rests on two established facts:

- each row of the file is the exact p^+ integral minus $T_{\rm row}\log\frac{\vee}{\Lambda}$ (`Finite_terms_exact.pdf`, Sec. 3), so $C_{\rm row} = P_{\rm row} - T_{\rm row}$;
- $\sum P_{\rm row} = \mathcal{B}$ and $\sum T_{\rm row} = \mathcal{J}$ (`Final_evolution_check.pdf`, exactly).

Consequence. With the corrected evolution, $\mathcal{N}\frac{1}{2}\log\frac{\vee}{\Lambda}(\mathcal{J}+\mathcal{B})$, the part $\frac{1}{2}\log\frac{\vee}{\Lambda}\sum C_{\rm row} = \mathcal{N}\frac{1}{2}\log\frac{\vee}{\Lambda}(\mathcal{B}-\mathcal{J})$ belongs to the evolution. So:

$$\boxed{\text{in every row: } C_{\rm row}\log\frac{\vee}{k^+} \;\longrightarrow\; C_{\rm row}\,L_*, \qquad L_* = \log\frac{\sqrt{\vee\Lambda}}{k^+}, \qquad \sum_{\rm rows} C_{\rm row}L_* = \mathcal{N}\log\frac{k^+}{\sqrt{\vee\Lambda}}(\mathcal{J}-\mathcal{B}).} \tag{6.2}$$

- The last expression is the finite remainder of Sec. 5 of `Evolution_terms_corrected.tex`. It is counted once: either through the rows below, or through that Sec. 5, not both.
- With your earlier convention ($\log\frac{\vee}{k^+}$ itself counted as finite, evolution $\mathcal{N}\log\frac{\vee}{\Lambda}\mathcal{J}$), read $L_* \to L_\vee$ everywhere below.

7 Step 5: the rows, simplified

Each row keeps its prefactor $\frac{1}{(2\pi)^3}$, its phase and its colour factor, as in your file. All p^+ integrals are done. Group I and Group III: +c.c. (Sec. 2.1).

7.1 Group I (each row + c.c.)

Row II. Part 1 and 2b UV: Secs. 4, 5. Parts 2a, 2b have the colour

$$\mathcal{O}_{\rm II} \equiv \mathcal{C}_L^a\Big[U^{ah}(w) f^{hbc}\{\rho^b(x),\rho^c(y)\} - f^{abc}U^{bd}(x)U^{ce}(y)\{\rho^d(x),\rho^e(y)\}\Big],$$

and

$$\begin{aligned}\frac{d^3N_{\rm II}}{d^3k} &= \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,x'} \mathcal{K}_{\rm II}\,\mathcal{O}_{\rm II} + \text{c.c.},\\ \mathcal{K}_{\rm II} \equiv& -\frac{ig^4}{32\pi^4 k^+}\,L_*\,\frac{(x'-w')^i}{(x'-w')^2}\Big[\frac{(x-w)^i}{(x-w)^2}\log\frac{(w-y)^2}{(x-y)^2} - \frac{(y-w)^i}{(y-w)^2}\log\frac{(w-x)^2}{(x-y)^2}\Big]\\ &- \frac{g^4}{64\pi^6 k^+}\int d^2p\,d^2k\,e^{ik\cdot(w-y)}e^{ip\cdot(y-x)}\frac{(x'-w')^i}{(x'-w')^2\,p^2k^2}\,R^i(k,p),\end{aligned} \tag{7.1}$$

$$R^i = -2k^i\log\frac{p^2}{k^2+a\,p^2} + \frac{(2p^2-2k\cdot p+k^2)k^i+k^2p^i}{(k-p)^2}\log\Big[\frac{k^2}{(k-p)^2}+a\Big(1-\frac{k^2}{(k-p)^2}\Big)\Big] - \frac{p^2k^i+k^2p^i}{(k-p)^2}\log(1-a).$$

2b has no large logarithm (it has a finite limit for $a\to 0$); it stays in momentum space.

Row III. All of N_1, N_2, N_{3a}, N_{3b}; colour $\mathcal{O}_{\rm III} = f^{abd}\mathcal{C}_L^a[U^{de}(x)U^{bc}(z)\rho^e(x)\rho^c(y) - U^{de}(x)U^{bc}(y)\rho^e(x)\rho^c(y)]$:

$$\frac{d^3N_{\rm III}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,z,x'} \mathcal{K}_{\rm III}\,\mathcal{O}_{\rm III} + \text{c.c.}, \tag{7.2}$$

$$\begin{aligned}\mathcal{K}_{\rm III} \equiv\, & p_8\,L_*\Big[(X'\cdot X_w)(Y\cdot Z_w - Y\cdot X_z) + \frac{X'\cdot Y}{(z-w)^2} - \Sigma_4\Big] + \lambda_2\Big[p_{16}T_1 - p_8(X'\cdot X_w)(X_z\cdot Y) + p_8(Q-\Sigma_4)\Big]\\ &- \log\big(1-a\big)\big[p_{16}T_1 + p_8 Q\big] + p_8\mathcal{A}_0 M_0 + N_{3b},\end{aligned}$$

$$\begin{aligned}T_1 &= (X'\cdot Y)\frac{A+B}{(A-B)^2}, \qquad Q = (X'\cdot Y)\,\frac{[(z-w)\cdot(x-z)]B - [(z-w)\cdot(x-w)]A}{(z-w)^2(B-A)^2},\\ \Sigma_4 &= \frac{(X'\cdot Z_w)\,Y\cdot(x-w) - (X'\cdot X_w)(Y\cdot Z_w)A - (X'\cdot Z_w)(Y\cdot X_z)B + X'\cdot(x-z)\,(Y\cdot Z_w)}{B-A},\\ M_0 &= (X'\cdot Z_w)\,Y\cdot(x-w) + (Y\cdot Z_w)\,X'\cdot(x-z) - \frac{(x-w)\cdot(x-z)}{A}\,(X'\cdot Z_w)\,Y\cdot(x-z),\\ N_{3b} &= -p_8\mathcal{A}_0\frac{(x-w)\cdot(x-z)}{B}(Y\cdot Z_w)\,X'\cdot(x-w)\\ &\quad + p_{16}\log\frac{(z-w)^2}{B}\Big[-\frac{(X'\cdot Z_w)\,Y\cdot(x-z)}{A} + \frac{(Y\cdot Z_w)\,X'\cdot(x-w)}{B} - \frac{X'\cdot Y}{(z-w)^2}\Big].\end{aligned}$$

Row IV. Colour $\mathcal{O}_{\rm IV} = \mathcal{C}_L^a[f^{cdc'}U^{ac}(w)U^{bd}(z)U^{be}(y)\rho^e(y)\rho^{c'}(x) - f^{abc}U^{be}(y)U^{cd}(x)\rho^e(y)\rho^d(x)]$:

$$\begin{aligned}\frac{d^3N_{\rm IV}}{d^3k} &= \frac{1}{(2\pi)^3}\int e^{-ik\cdot(w'-w)}\,p_8\,\mathcal{K}_{\rm IV}\,\mathcal{O}_{\rm IV} + \text{c.c.},\\ \mathcal{K}_{\rm IV} &\equiv \Big[X'^iY^j\mathbb{T}^{ij} - \frac{X'\cdot Y}{2(z-w)^2}\Big]L_* + X'^iY^j\mathbb{T}^{ij}\lambda + \log\big(1-a\big)\,X'^iY^j\mathbb{L}^{ij}.\end{aligned} \tag{7.3}$$

Row V, part 1. $\beta \equiv k\cdot(z-x)$; $\kappa_a = [X'\cdot\mathcal{K}(z-x)][Y\cdot\mathcal{K}(y'-x)]$, $\kappa_b = [\mathcal{K}(y'-x)\cdot\mathcal{K}(z-x)](X'\cdot Y)$, $\kappa_q = [X'\cdot\mathcal{K}(y'-x)][\mathcal{K}(z-x)\cdot Y]$:

$$\begin{aligned}\frac{d^3N_{\rm V1}}{d^3k} = \frac{1}{(2\pi)^3}\frac{ig^4}{32\pi^5k^+}\int_{x,z,y,x',y',w'}&\Big\{e^{-ik\cdot(w'-z)}\Big[\kappa_a\,\frac{1-e^{-i\beta}}{i\beta} + \kappa_b\big(\mathrm{Ein}(i\beta)+L_*\big)\Big]\\ &+ e^{-ik\cdot(w'-x)}\,\kappa_q\big(\mathrm{Ein}(-i\beta)+L_*\big)\Big\}\,\mathcal{O}_{\rm V1} + \text{c.c.},\\ \mathcal{O}_{\rm V1} \equiv \mathcal{C}_L^a\Big[&f^{adc}U^{db}(x)U^{ce}(z)\{\rho^e(y),\rho^b(y')\} - f^{abc}U^{cd}(y)U^{be}(y')\{\rho^d(y),\rho^e(y')\}\Big].\end{aligned} \tag{7.4}$$

Row V, part 2 (2a, 2b finite). $Y_x \equiv (y-x)^2$, $Y_z \equiv (y-z)^2$, $\delta \equiv Y_x - Y_z$, $c \equiv Y_z/\delta$, $m_0 = \frac{1-e^{-i\beta}}{i\beta}$, $m_1 = \frac{m_0 - e^{-i\beta}}{i\beta}$; master integrals

$$\mathcal{M}_0 = \frac{e^{i\beta c}}{\delta}\big[E_1(i\beta c) - E_1(i\beta(1+c))\big], \quad \mathcal{M}_1 = \frac{m_0}{\delta} - c\,\mathcal{M}_0, \quad \mathcal{M}_2 = \frac{m_1 - c\,m_0}{\delta} + c^2\mathcal{M}_0,$$

($\mathcal{M}_n = \int_0^1 dt \frac{t^n e^{-i\beta t}}{tY_x+(1-t)Y_z}$), $\mathcal{N}_0 = \mathcal{M}_0$ with $\beta \to -\beta$, $Y_x \leftrightarrow Y_z$. Your brackets contracted with $X'^i \mathcal{K}^m(z-x)$:

$$
\begin{aligned}
s_1 &= X'\cdot \mathcal{K}(z-x)\,\frac{d_\perp}{2(x-z)^2}(Y_x - Y_z),\\
v_2 &= X'\cdot \mathcal{K}(z-x)\Big[\frac{(y-z)\cdot(x-z)}{(x-z)^2} + \frac{(y-x)\cdot(y-z)}{2Y_x} - \frac{Y_x-Y_z}{2(x-z)^2}\Big],\\
v_4 &= X'\cdot \mathcal{K}(z-x)\Big[\frac{(y-x)\cdot(x-z)}{(x-z)^2} - \frac{(y-x)\cdot(y-z)}{2Y_z} - \frac{Y_x-Y_z}{2(x-z)^2}\Big],\\
s_3 &= \frac{X'\cdot(y-z)\,\mathcal{K}(z-x)\cdot(x-z)}{(x-z)^2} + \frac{\mathcal{K}(z-x)\cdot(y-x)\,X'\cdot(y-z)}{2Y_x},\\
s_5 &= \frac{\mathcal{K}(z-x)\cdot(y-x)\,X'\cdot(x-z)}{(x-z)^2} - \frac{\mathcal{K}(z-x)\cdot(y-x)\,X'\cdot(y-z)}{2Y_z}\\
&\quad + \frac{\mathcal{K}(z-x)\cdot(y-z)\,X'\cdot(x-z)}{(x-z)^2} + \frac{X'\cdot(y-x)\,\mathcal{K}(z-x)\cdot(y-z)}{2Y_x},\\
s_6 &= \frac{X'\cdot(y-x)\,\mathcal{K}(z-x)\cdot(x-z)}{(x-z)^2} - \frac{X'\cdot(y-x)\,\mathcal{K}(z-x)\cdot(y-z)}{2Y_z}.
\end{aligned}
$$

Then ($d_\perp = 2$)

$$
\begin{aligned}
\{\dots\}_{\mathrm{V2}} \equiv e^{-ik\cdot(w'-z)}&\Big[s_1(\mathcal{M}_1-\mathcal{M}_2) + v_2(\mathcal{M}_0-\mathcal{M}_1) + v_4\mathcal{M}_1 + \Big(s_3\frac{Y_x}{Y_z} - s_5\Big)\mathcal{M}_0 + \frac{s_3}{Y_z}\big(\mathrm{Ein}(i\beta)+L_*\big)\Big]\\
&+ e^{-ik\cdot(w'-x)}\,s_6\Big[\frac{Y_z}{Y_x}\mathcal{N}_0 + \frac{1}{Y_x}\big(\mathrm{Ein}(-i\beta)+L_*\big)\Big],
\end{aligned}
\tag{7.5}
$$

$$
\begin{aligned}
\frac{d^3N_{2a}}{d^3k} &= \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5k^+}\int_{x,y,z,x',w'}\{\dots\}_{\mathrm{V2}}\\
&\quad\times\big[U^{ab'}(x') - U^{ab'}(w')\big]\big[-f^{adc}f^{fbe}U^{db}(x)U^{ce}(z) + N_cU^{af}(x)\big]\rho^{b'}(x')\rho^f(y) + \text{c.c.},
\end{aligned}
\tag{7.6}
$$

$$
\begin{aligned}
\frac{d^3N_{2b,\mathrm{fin}}}{d^3k} &= \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5k^+}\int_{x,y,z,x',w'}\Big\{\{\dots\}_{\mathrm{V2}}\\
&\quad\times\big[U^{ab'}(x') - U^{ab'}(w')\big]N_c\big[U^{af}(y) - U^{af}(x)\big]\rho^{b'}(x')\rho^f(y)\\
&\quad + e^{-ik\cdot(w'-z)}\Big(\frac{11}{6}+2L_*\Big)\frac{X'\cdot Y\,(y-x)\cdot(y-z)}{(y-x)^2(x-z)^2}\\
&\quad\times\big[U^{ab'}(x') - U^{ab'}(w')\big]N_c\big[U^{af}(y) - U^{af}(z)\big]\rho^{b'}(x')\rho^f(y)\Big\} + \text{c.c.}
\end{aligned}
\tag{7.7}
$$

7.2 Group II (no c.c.)

Row I, part 1. Colour of block A:

$$
\mathcal{O}_A \equiv \big[U^{ac'}(w')\{\rho^{c'}(x'),\rho^b(y)\} - U^{bc'}(z)U^{ad'}(x')U^{c'e'}(y)\{\rho^{d'}(x'),\rho^{e'}(y)\}\big]\mathcal{R}^{ab}(w,z,x).
$$

$$
\frac{d^3N_{\mathrm{I}(1)}}{d^3k} = \frac{1}{(2\pi)^3}\int e^{-ik\cdot(w'-w)}\Big[-p_{16}\,\mathcal{K}_{\mathrm{IV}}\,\mathcal{O}_A\Big] + \text{block B}, \qquad \text{block B} = \Big[-p_{16}\,\mathcal{K}_{\mathrm{IV}}\,\mathcal{O}_A\Big]^*_{w\leftrightarrow w'}\ \text{(checked)}. \tag{7.8}
$$

Row I, part 2 (2a–2f), collected. With your brackets (all from the file)

$$
\begin{aligned}
[\dots]_{2a} &= \frac{(x-w)\cdot(z-w')\,(x'-z)\cdot(z-w)}{(z-w)^2(z-w')^2} - \frac{(x-w)\cdot(x'-w')\,(z-w)\cdot(x'-z)}{2(z-w)^2B'}\\
&\quad + \frac{(x-z)\cdot(x'-z)\,(x-w)\cdot(z-w')}{2A(z-w')^2} - \frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(x-w)}{4AB'}\\
&\quad + \frac{(x'-w')\cdot(z-w)\,(x-z)\cdot(z-w')}{(z-w)^2(z-w')^2} + \frac{(x'-w')\cdot(z-w)\,(x-z)\cdot(x'-z)}{2(z-w)^2A'}\\
&\quad - \frac{(x-z)\cdot(z-w')\,(x'-w')\cdot(x-w)}{2B(z-w')^2} - \frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(x-z)}{4BA'},\\
[\dots]_{2b} &= \frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2(z-w')^2} + \frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2(z-w)^2A'}\\
&\quad + \frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2A(z-w')^2} + \frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4AA'},
\end{aligned}
$$

$$
\begin{aligned}
[\dots]_{2c} &= \frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2(z-w')^2} - \frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2(z-w)^2B'} \\
&\quad - \frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2B(z-w')^2} + \frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4BB'}, \\
[\dots]_{2d} &= \frac{(x-w)\cdot(z-w)\,(A'-B')}{2(z-w)^2(z-w')^2} + \frac{(x-z)\cdot(x-w)\,(A'-B')}{4A(z-w')^2} \\
&\quad + \frac{(x'-w')\cdot(z-w')\,(A-B)}{2(z-w)^2(z-w')^2} + \frac{(x'-z)\cdot(x'-w')\,(A-B)}{4(z-w)^2A'}, \\
[\dots]_{2e} &= \frac{(x-z)\cdot(z-w)\,(A'-B')}{2(z-w)^2(z-w')^2} - \frac{(x-z)\cdot(x-w)\,(A'-B')}{4B(z-w')^2} \\
&\quad + \frac{(x'-z)\cdot(z-w')\,(A-B)}{2(z-w)^2(z-w')^2} - \frac{(x'-z)\cdot(x'-w')\,(A-B)}{4(z-w)^2B'},
\end{aligned}
$$

and your colour $\mathcal{O}_{58} = f^{c'bd'}f^{cbd}U^{ac'}(w')U^{ac}(w)\rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}f^{acd}U^{ac'}(w')U^{bc}(z)U^{de}(x)\rho^{d'}(x')\rho^{e}(x) - f^{ac'd'}f^{cbd}U^{ac}(w)U^{bc'}(z)U^{e'd'}(x')\rho^{e'}(x')\rho^{d}(x) + N_cU^{e'd}(x')U^{de}(x)\rho^{e'}(x')\rho^{e}(x)$:

$$
\boxed{\frac{d^3N_{\mathrm{I}(2)}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^5k^+}\int_{z,w,w',x,x'} e^{-ik\cdot(w'-w)}\Big[c_\lambda\,\lambda + c_{\lambda'}\,\lambda' + c_L\,L_* + c_\ell\,\log(1-a) - \frac{d_\perp}{4}\,\frac{1-a}{(z-w)^2(z-w')^2}\Big]\mathcal{O}_{58},}
\tag{7.9}
$$

$$
\begin{aligned}
c_\lambda &= \frac{1}{D}\Big[[\dots]_{2a} - \frac{A}{B}[\dots]_{2b} - \frac{B}{A}[\dots]_{2c} + \frac{A[\dots]_{2d} - B[\dots]_{2e}}{A-B} - \frac{d_\perp}{4}\,\frac{AB(A'-B')}{(A-B)(z-w)^2(z-w')^2}\Big], \\
c_{\lambda'} &= \frac{1}{D}\Big[-[\dots]_{2a} + \frac{A'}{B'}[\dots]_{2b} + \frac{B'}{A'}[\dots]_{2c} + \frac{A'[\dots]_{2d} - B'[\dots]_{2e}}{B'-A'} + \frac{d_\perp}{4}\,\frac{A'B'(A-B)}{(A'-B')(z-w)^2(z-w')^2}\Big], \\
c_L &= -\frac{[\dots]_{2b}}{BB'} + \frac{[\dots]_{2c}}{AA'}, \qquad c_\ell = \frac{[\dots]_{2b}}{BB'}.
\end{aligned}
$$

- The explicit $\log\frac{\vee}{k^+ + \Lambda}$ of 2d, 2e, 2f cancel against the $L_\vee$ inside the other logarithms.
- The $\log(1-a)$ of 2b becomes c_ℓ.
- Eq. (7.9) vs. your 2a–2f: 1.2×10^{-15}.

Rows II, III. Four ρ only (Sec. 10). **Row IV:** Sec. 5.

7.3 Group III (each row + c.c.)

Rows I, II. $-p_8\,\mathcal{K}_{\mathrm{IV}}$ with the colour of the row:

$$
\frac{d^3N_{\mathrm{III,I}}}{d^3k} = \frac{1}{(2\pi)^3}\int e^{-ik\cdot(w'-w)}\,(-p_8)\mathcal{K}_{\mathrm{IV}}\,\mathcal{O}_{\mathrm{G3I}} + \text{c.c.}, \qquad \frac{d^3N_{\mathrm{III,II}}}{d^3k} = \frac{1}{(2\pi)^3}\int e^{-ik\cdot(w'-w)}\,(-p_8)\mathcal{K}_{\mathrm{IV}}\,\mathcal{O}_{\mathrm{G3II}} + \text{c.c.},
\tag{7.10}
$$

with $\mathcal{O}_{\mathrm{G3I}} = [U^{ab'}(x')U^{c'd'}(y)U^{c'b}(z)\rho^{b'}(x')\rho^{d'}(y) - U^{bd'}(z)U^{d'e'}(y)U^{ac'}(w')\rho^{c'}(x')\rho^{e'}(y)]\mathcal{R}^{ab}(w,z,x)$ and $\mathcal{O}_{\mathrm{G3II}} = [U^{c'd'}(y)U^{ab'}(x')U^{bc'}(z)\rho^{d'}(y)\rho^{b'}(x') - U^{ab'}(x')\rho^{b}(y)\rho^{b'}(x')]\mathcal{R}^{ab}(w,z,x)$.

Rows III part 1, V, VI. $\kappa \equiv k\cdot(y'-z)$, $t_1 = [\mathcal{K}(y'-z)\cdot\mathcal{K}(x'-y')][\mathcal{K}(y-w)\cdot\mathcal{K}(x-z)]$, $t_2 = [\mathcal{K}(y'-z)\cdot\mathcal{K}(x-z)][\mathcal{K}(y-w)\cdot\mathcal{K}(x'-y')]$, $t_3 = [\mathcal{K}(y'-z)\cdot\mathcal{K}(y-w)][\mathcal{K}(x-z)\cdot\mathcal{K}(x'-y')]$:

$$
\begin{aligned}
\mathcal{K}_C &\equiv t_1\,e^{i\kappa}\Big[E_1(i\kappa) - E_1\Big(\frac{i\kappa}{a}\Big)\Big] - t_3\,\frac{1-e^{-i\kappa\frac{1-a}{a}}}{i\kappa} + t_2\Big[\gamma_E + \log(i\kappa) + E_1\Big(i\kappa\frac{1-a}{a}\Big)\Big] + t_2\,L_*, \\
\frac{d^3N_C}{d^3k} &= \frac{1}{(2\pi)^3}\frac{c_C}{k^+}\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,\mathcal{K}_C\,\mathcal{O}_C + \text{c.c.},
\end{aligned}
\tag{7.11}
$$

with $c_{\mathrm{III}(1)} = \frac{ig^4}{16\pi^5}$, $c_{\mathrm{V}} = c_{\mathrm{VI}} = \frac{ig^4}{8\pi^5}$ and the colour factors of the file:

$$
\begin{aligned}
\mathcal{O}_{\mathrm{III}(1)} &= f^{c'd'a}\big[U^{e'c'}(y')U^{d'b}(z) - U^{bd'}(z)U^{c'e'}(x')\big]\rho^{e'}(x') \\
&\quad\times\big[U^{ac}(w)\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)U^{ad}(y)U^{ce}(x)\{\rho^d(y),\rho^e(x)\}\big], \\
\mathcal{O}_{\mathrm{V}} &= f^{c'ca}\big[U^{e'c'}(y') - U^{c'e'}(x')\big]\rho^{e'}(x')\big[U^{ce}(x)U^{ad}(y) - U^{ad}(w)U^{ce}(x)\big]\rho^{e}(x)\rho^{d}(y), \\
\mathcal{O}_{\mathrm{VI}} &= f^{c'd'a}\big[U^{e'c'}(y')U^{d'b}(z) - U^{bd'}(z)U^{c'e'}(x')\big]\rho^{e'}(x')\big[U^{bc}(z)U^{ad}(y)U^{ce}(x)\rho^{d}(y)\rho^{e}(x) - U^{ad}(y)\rho^{d}(y)\rho^{b}(x)\big].
\end{aligned}
$$

For $y'\to z$ ($\kappa\to0$) the brace is finite: $t_1L_\vee - t_3\frac{1-a}{a} - t_2\log(1-a) + t_2(L_* - L_\vee)$. The term $e^{-i\kappa\frac{1-a}{a}}$ is bounded and gives no logarithm after $\int_{y'}$.

Row III part 2. $b \equiv B/A$, $U_a \equiv \frac{1-a}{a}$, $\mathcal{J}(b) \equiv e^{i\kappa b}[E_1(i\kappa b) - E_1(i\kappa(b+U_a))]$, $\mathcal{J}_0 \equiv \frac{1-e^{-i\kappa U_a}}{i\kappa}$, $\mathcal{G} \equiv \gamma_E + \log(i\kappa) + E_1(i\kappa U_a)$; with $D_p = p^+A + k^+B$, $P = p^+ + k^+$:

$$\int_0^{\vee - k^+} dp^+ e^{-i\kappa \frac{p^+}{k^+}} \Big\{ \frac{p^+}{P^2 D_p}, \frac{p^+}{P D_p}, \frac{1}{P D_p}, \frac{p^+}{D_p}, \frac{1}{D_p} \Big\} \equiv \{\mathbb{I}_1, \ldots, \mathbb{I}_5\}$$
$$= \Big\{ \frac{1}{Ak^+} \Big[\frac{b\,(\mathcal{J}(1) - \mathcal{J}(b))}{(1-b)^2} + \frac{1 - a\,e^{-i\kappa U_a} - i\kappa \mathcal{J}(1)}{1-b} \Big],\ \frac{\mathcal{J}(1) - b\,\mathcal{J}(b)}{A(1-b)},\ \frac{\mathcal{J}(1) - \mathcal{J}(b)}{k^+ A(b-1)},\ \frac{k^+}{A}\big(\mathcal{J}_0 - b\,\mathcal{J}(b)\big),\ \frac{\mathcal{J}(b)}{A} \Big\}.$$

Your bracket is $\frac{1}{D_p}\big[c_1 \frac{p^+}{P^2} + c_2 \frac{p^+}{k^+P} + c_3 \frac{1}{P} + c_4 \frac{p^+}{k^{+2}} + c_5 \frac{1}{k^+}\big]$ plus the $[\frac{1}{p^+}]_+$ term with coefficient s, where ($u \equiv \mathcal{K}(y'-z)$, $v \equiv \mathcal{K}(x'-y')$):

- $c_1 = (u\cdot v)\frac{d_\perp (A-B)}{2(z-w)^2}$;
- $c_2 = (u\cdot v)\big[\frac{(x-z)\cdot(z-w)}{(z-w)^2} - \frac{(x-z)\cdot(x-w)}{2B} - \frac{A-B}{2(z-w)^2}\big]$;
- $c_3 = (u\cdot v)\big[\frac{(x-w)\cdot(z-w)}{(z-w)^2} + \frac{(x-z)\cdot(x-w)}{2A} - \frac{A-B}{2(z-w)^2}\big]$;
- $c_4 = -\frac{(v\cdot(x-z))(u\cdot(z-w))}{(z-w)^2} + \frac{(v\cdot(x-z))(u\cdot(x-w))}{2B}$;
- $c_5 = -\big[\frac{(u\cdot(x-z))(v\cdot(z-w))}{(z-w)^2} - \frac{(u\cdot(x-z))(v\cdot(x-w))}{2B} + \frac{(u\cdot(x-w))(v\cdot(z-w))}{(z-w)^2} + \frac{(v\cdot(x-z))(u\cdot(x-w))}{2A}\big]$;
- $s = -\frac{(v\cdot(x-w))(u\cdot(z-w))}{(z-w)^2} - \frac{(u\cdot(x-z))(v\cdot(x-w))}{2A}$.

Then

$$\frac{d^3 N_{\mathrm{III}(2)}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^5}\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)} \Big\{ c_1\mathbb{I}_1 + \frac{c_2}{k^+}\mathbb{I}_2 + c_3\mathbb{I}_3 + \frac{c_4}{k^{+2}}\mathbb{I}_4 + \frac{c_5}{k^+}\mathbb{I}_5 - \frac{s}{k^+ B}\big[\mathcal{G} + L_* + \mathcal{J}(b)\big] \Big\} \mathcal{O}_{\mathrm{III}(2)} + \text{c.c.}, \tag{7.12}$$

with $\mathcal{O}_{\mathrm{III}(2)} = f^{c'd'a}\big[U^{e'c'}(y')U^{d'b}(z) - U^{bd'}(z)U^{c'e'}(x')\big]\rho^{e'}(x')\,\mathcal{R}^{ab}(w,z,x)$ (two ρ's).

8 Step 6: rows with the same kernel

The $\mathcal{K}_{\mathrm{IV}}$ family. Group I Row IV, Group III Rows I, II and Group II Row I part 1 all have the kernel $\mathcal{K}_{\mathrm{IV}}$ of Eq. (7.3). Their sum, with the c.c.'s, is

$$\frac{d^3 N_{\mathcal{K}_{\mathrm{IV}}}}{d^3k} = \frac{1}{(2\pi)^3}\int e^{-ik\cdot(w'-w)}\, p_8\, \mathcal{K}_{\mathrm{IV}} \Big[\mathcal{O}_{\mathrm{IV}} - \mathcal{O}_{\mathrm{G3I}} - \mathcal{O}_{\mathrm{G3II}} - \tfrac{1}{2}\mathcal{O}_A\Big] + \text{c.c.} \tag{8.1}$$

The c.c. of the $\frac{1}{2}\mathcal{O}_A$ term is Group II block B.

- **Symmetric (classical) part** (4.0×10^{-15}):

$$\mathcal{O}_{\mathrm{IV}} - \mathcal{O}_{\mathrm{G3I}} - \mathcal{O}_{\mathrm{G3II}} - \tfrac{1}{2}\mathcal{O}_A \xrightarrow{S} \boxed{f^{acd}\,\mathcal{C}^a_L \Big[\big(U(w)\rho(y)\big)^c \big(U(w)\rho(x)\big)^d - \big(U(z)\rho(y)\big)^c \big(U(x)\rho(x)\big)^d\Big]} \tag{8.2}$$

- In particular, $\mathcal{O}_{\mathrm{IV}}$ and $\mathcal{O}_{\mathrm{G3I}}$ have the same symmetric part (3.6×10^{-15}), so Group I Row IV and Group III Row I cancel there.
- What remains is $-\mathcal{O}_{\mathrm{G3II}} - \frac{1}{2}\mathcal{O}_A = \mathcal{C}^a_L \rho^b(y)\mathcal{R}^{ab}(w,z,x)$ (4.0×10^{-16}).
- The two-ρ parts of the four rows do not cancel. They are in Sec. 9.

The $\mathcal{K}_C$ family. Group III Rows III(1), V, VI have the kernel $\mathcal{K}_C$ of Eq. (7.11):

$$\frac{d^3 N_{\mathcal{K}_C}}{d^3k} = \frac{1}{(2\pi)^3}\frac{ig^4}{16\pi^5 k^+}\int e^{-ik\cdot(y'-w)}\, \mathcal{K}_C \Big[\mathcal{O}_{\mathrm{III}(1)} + 2\mathcal{O}_{\mathrm{V}} + 2\mathcal{O}_{\mathrm{VI}}\Big] + \text{c.c.} \tag{8.3}$$

- **Symmetric part** (1.4×10^{-14}):

$$\mathcal{O}_{\mathrm{III}(1)} + 2\mathcal{O}_{\mathrm{V}} + 2\mathcal{O}_{\mathrm{VI}} \xrightarrow{S} \boxed{2f^{abc}\Big[\big(U(y') - U(x')\big)\rho(x')\Big]^a \Big[\big(U(z) - U(x)\big)\rho(x)\Big]^b \Big[\big(U(w) - U(y)\big)\rho(y)\Big]^c} \tag{8.4}$$

  Each factor is a difference of Wilson lines, like the LO amplitude.
- The $U(z)U(y)U(x)$ terms of Rows III(1) and VI cancel.

9 Step 7: three ρ's = symmetric three-ρ part + two-ρ part

Rule. Write the colour operator of a row with the charges in your order, $\mathcal{O} = T^{a_1 a_2 a_3}\rho^{a_1}(u_1)\rho^{a_2}(u_2)\rho^{a_3}(u_3)$, with $\{B,C\} = BC + CB$ written out. With $[\rho^a(u), \rho^b(v)] = if^{abc}\rho^c(u)\delta(u-v)$,

$$\mathcal{O} = S(\mathcal{O}) + \mathcal{O}^{(2)}, \qquad \mathcal{O}^{(2)} = \frac{1}{2}\sum_{i<j} \mathcal{O}\Big|_{\rho^{a_i}(u_i)\rho^{a_j}(u_j)\to if^{a_i a_j g}\rho^g(u_i)\,\delta(u_i-u_j)}. \tag{9.1}$$

S is the fully symmetric (classical) part. For $A\{B,C\}$ this gives $[A,B]C + [A,C]B$; for $\{B,C\}A$ it gives $[B,A]C + [C,A]B$.

The kernel of a row is a function of the charge points. So

$$\text{row} = \frac{1}{(2\pi)^3}\int \text{phase}\, \mathcal{K}_{\text{row}}\, S(\mathcal{O}_{\text{row}}) \;+\; \frac{1}{(2\pi)^3}\int \text{phase} \sum_{\text{merges}} \mathcal{K}_{\text{row}}\big|_{u_j\to u_i}\, \mathcal{O}^{(2)}_{\text{row}}\ (\text{+c.c.}).$$

- The symmetric parts are the classical first saturation correction of the projectile. They vanish for a Gaussian projectile.
- The two-ρ parts below were generated with Eq. (9.1), simplified ($f^{abr}f^{abs} = N_c\delta^{rs}$, $U^{ij}(p)U^{ik}(p) = \delta^{jk}$, differences $[U(p) - U(q)]$ collected), and checked against the reordering engine that reproduced your Sec. 2 lists exactly (agreement $\le 1.9\times 10^{-15}$).

Symmetric parts (checked, $\le 2.9\times 10^{-14}$):

$$\begin{aligned}
S(\mathcal{O}_{\text{II}}) &= 2f^{abc}\mathcal{C}^a_L\big[(U(w)\rho(x))^b(U(w)\rho(y))^c - (U(x)\rho(x))^b(U(y)\rho(y))^c\big],\\
S(\mathcal{O}_{\text{III}}) &= f^{abd}\mathcal{C}^a_L\big[(U(z)-U(y))\rho(y)\big]^b\big(U(x)\rho(x)\big)^d,\\
S(\mathcal{O}_{\text{V1}}) &= 2f^{abc}\mathcal{C}^a_L\big[(U(x)\rho(y'))^b(U(z)\rho(y))^c - (U(y')\rho(y'))^b(U(y)\rho(y))^c\big],
\end{aligned}$$

and Eqs. (8.2), (8.4) for the two families.

Two-ρ parts. Notation: $\mathcal{K}|_{x\to x'}$ is the kernel with x set equal to x', etc.; the colour is in your index notation.

Group I, Row II (2a, 2b), colour $\mathcal{O}_{\text{II}}$ (+c.c.).

$$\begin{aligned}
\mathcal{O}^{(2)}\text{-part} = \mathcal{K}_{\text{II}}\big|_{x\to x'}\Big[&- i\, f^{hbc}\, f^{bb'g}\,\big[U^{ab'}(x') - U^{ab'}(w')\big]\,U^{ah}(w)\,\rho^g(x')\,\rho^c(y)\\
&- i\, f^{abc}\, f^{b'dg}\,\big[U^{ab'}(x') - U^{ab'}(w')\big]\,U^{bd}(x')\,U^{ce}(y)\,\rho^g(x')\,\rho^e(y)\Big]\\
+\,\mathcal{K}_{\text{II}}\big|_{y\to x'}\Big[&\,i\, f^{hbc}\, f^{b'cg}\,\big[U^{ab'}(x') - U^{ab'}(w')\big]\,U^{ah}(w)\,\rho^g(x')\,\rho^b(x)\\
&- i\, f^{abc}\, f^{b'eg}\,\big[U^{ab'}(x') - U^{ab'}(w')\big]\,U^{bd}(x)\,U^{ce}(x')\,\rho^g(x')\,\rho^d(x)\Big]
\end{aligned} \tag{9.2}$$

Group I, Row III, colour $\mathcal{O}_{\text{III}}$ (+c.c.).

$$\begin{aligned}
\mathcal{O}^{(2)}\text{-part} = \mathcal{K}_{\text{III}}\big|_{x\to x'}\Big[&\tfrac{i}{2}\, f^{abd}\, f^{b'eg}\,\big[U^{ab'}(x') - U^{ab'}(w')\big]\,U^{de}(x')\,\big[U^{bc}(z) - U^{bc}(y)\big]\,\rho^g(x')\,\rho^c(y)\Big]\\
+\,\mathcal{K}_{\text{III}}\big|_{y\to x'}\Big[&\tfrac{i}{2}\, f^{abd}\, f^{b'cg}\,\big[U^{ab'}(x') - U^{ab'}(w')\big]\,U^{de}(x)\,\big[U^{bc}(z) - U^{bc}(x')\big]\,\rho^g(x')\,\rho^e(x)\Big]\\
+\,\mathcal{K}_{\text{III}}\big|_{y\to x}\Big[&-\tfrac{i}{2}\, f^{abd}\, f^{ceg}\,\big[U^{ab'}(x') - U^{ab'}(w')\big]\,U^{de}(x)\,\big[U^{bc}(z) - U^{bc}(x)\big]\,\rho^{b'}(x')\,\rho^g(x)\Big]
\end{aligned} \tag{9.3}$$

Group I, Row IV, colour $\mathcal{O}_{\text{IV}}$ (+c.c.).

$$\begin{aligned}
\mathcal{O}^{(2)}\text{-part} = p_8\mathcal{K}_{\text{IV}}\big|_{y\to x'}\Big[&\tfrac{i}{2}\, f^{cdc'}\, f^{b'eg}\,\big[U^{ab'}(x') - U^{ab'}(w')\big]\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(x')\,\rho^g(x')\,\rho^{c'}(x)\\
&- \tfrac{i}{2}\, f^{abc}\, f^{b'eg}\,\big[U^{ab'}(x') - U^{ab'}(w')\big]\,U^{be}(x')\,U^{cd}(x)\,\rho^g(x')\,\rho^d(x)\Big]\\
+\,p_8\mathcal{K}_{\text{IV}}\big|_{x\to x'}\Big[&\tfrac{i}{2}\, f^{cdc'}\, f^{b'c'g}\,\big[U^{ab'}(x') - U^{ab'}(w')\big]\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^g(x')\,\rho^e(y)\\
&- \tfrac{i}{2}\, f^{abc}\, f^{b'dg}\,\big[U^{ab'}(x') - U^{ab'}(w')\big]\,U^{be}(y)\,U^{cd}(x')\,\rho^g(x')\,\rho^e(y)\Big]\\
+\,p_8\mathcal{K}_{\text{IV}}\big|_{y\to x}\Big[&-\tfrac{i}{2}\, f^{cdc'}\, f^{c'eg}\,\big[U^{ab'}(x') - U^{ab'}(w')\big]\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(x)\,\rho^{b'}(x')\,\rho^g(x)\\
&+ \tfrac{i}{2}\, f^{abc}\, f^{deg}\,\big[U^{ab'}(x') - U^{ab'}(w')\big]\,U^{be}(x)\,U^{cd}(x)\,\rho^{b'}(x')\,\rho^g(x)\Big]
\end{aligned} \tag{9.4}$$

Group I, Row V part 1, colour $\mathcal{O}_{\rm V1}$ (+c.c.); $\mathcal{K}_{\rm V1}$ is the whole of Eq. (7.4) before $\mathcal{O}_{\rm V1}$, with its prefactor $\frac{ig^4}{32\pi^5 k^+}$ and its two phases.

$$\begin{aligned}
\mathcal{O}^{(2)}\text{-part} = \mathcal{K}_{\rm V1}\big|_{y\to x'}\Big[& i\, f^{adc}\, f^{b'eg}\left[U^{ab'}(x') - U^{ab'}(w')\right] U^{db}(x)\, U^{ce}(z)\, \rho^g(x')\, \rho^b(y') \\
& - i\, f^{abc}\, f^{b'dg}\left[U^{ab'}(x') - U^{ab'}(w')\right] U^{cd}(x')\, U^{be}(y')\, \rho^g(x')\, \rho^e(y')\Big] \\
+ \mathcal{K}_{\rm V1}\big|_{y'\to x'}\Big[& - i\, f^{adc}\, f^{bb'g}\left[U^{ab'}(x') - U^{ab'}(w')\right] U^{db}(x)\, U^{ce}(z)\, \rho^g(x')\, \rho^e(y) \\
& - i\, f^{abc}\, f^{b'eg}\left[U^{ab'}(x') - U^{ab'}(w')\right] U^{cd}(y)\, U^{be}(x')\, \rho^g(x')\, \rho^d(y)\Big]
\end{aligned} \tag{9.5}$$

Group II, Row I part 1, block A (no c.c.; block B is $[\dots]^*_{w\leftrightarrow w'}$ of this).

$$\begin{aligned}
\mathcal{O}^{(2)}\text{-part} = (-p_{16})\mathcal{K}_{\rm IV}\big|_{x\to x'}\Big[& i\, f^{cbd}\, f^{c'dg}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^g(x')\, \rho^b(y) \\
& - i\, f^{acd}\, f^{c'eg}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x')\, \rho^g(x')\, \rho^b(y) \\
& + i\, f^{cbd}\, f^{dd'g}\, U^{bc'}(z)\, U^{ad'}(x')\, U^{c'e'}(y)\, U^{ac}(w)\, \rho^g(x')\, \rho^{e'}(y) \\
& + i\, f^{ac'd}\, f^{d'eg}\, U^{ad'}(x')\, U^{c'e'}(y)\, U^{de}(x')\, \rho^g(x')\, \rho^{e'}(y)\Big] \\
+ (-p_{16})\mathcal{K}_{\rm IV}\big|_{y\to x}\Big[& - i\, f^{acd}\, f^{beg}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{c'}(x')\, \rho^g(x) \\
& + i\, f^{cbd}\, f^{de'g}\, U^{bc'}(z)\, U^{ad'}(x')\, U^{c'e'}(x)\, U^{ac}(w)\, \rho^{d'}(x')\, \rho^g(x) \\
& - i\, f^{ac'd}\, f^{ee'g}\, U^{ad'}(x')\, U^{c'e'}(x)\, U^{de}(x)\, \rho^{d'}(x')\, \rho^g(x) \\
& + i\, N_c\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{c'}(x')\, \rho^c(x)\Big]
\end{aligned} \tag{9.6}$$

Group III, Row I, colour $\mathcal{O}_{\rm G3I}$ (+c.c.).

$$\begin{aligned}
\mathcal{O}^{(2)}\text{-part} = (-p_8)\mathcal{K}_{\rm IV}\big|_{y\to x'}\Big[& \tfrac{i}{2}\, f^{cbd}\, f^{b'd'g}\, U^{ab'}(x')\, U^{c'd'}(x')\, U^{c'b}(z)\, U^{ac}(w)\, \rho^g(x')\, \rho^d(x) \\
& - \tfrac{i}{2}\, f^{acd}\, f^{b'd'g}\, U^{ab'}(x')\, U^{c'd'}(x')\, U^{c'b}(z)\, U^{bc}(z)\, U^{de}(x)\, \rho^g(x')\, \rho^e(x) \\
& - \tfrac{i}{2}\, f^{cbd}\, f^{c'e'g}\, U^{bd'}(z)\, U^{d'e'}(x')\, U^{ac'}(w')\, U^{ac}(w)\, \rho^g(x')\, \rho^d(x) \\
& + \tfrac{i}{2}\, f^{ad'd}\, f^{c'e'g}\, U^{d'e'}(x')\, U^{ac'}(w')\, U^{de}(x)\, \rho^g(x')\, \rho^e(x)\Big] \\
+ (-p_8)\mathcal{K}_{\rm IV}\big|_{x\to x'}\Big[& \tfrac{i}{2}\, f^{cbd}\, f^{b'dg}\, U^{ab'}(x')\, U^{c'd'}(y)\, U^{c'b}(z)\, U^{ac}(w)\, \rho^g(x')\, \rho^{d'}(y) \\
& - \tfrac{i}{2}\, f^{acd}\, f^{b'eg}\, U^{ab'}(x')\, U^{c'd'}(y)\, U^{c'b}(z)\, U^{bc}(z)\, U^{de}(x')\, \rho^g(x')\, \rho^{d'}(y) \\
& - \tfrac{i}{2}\, f^{cbd}\, f^{c'dg}\, U^{bd'}(z)\, U^{d'e'}(y)\, U^{ac'}(w')\, U^{ac}(w)\, \rho^g(x')\, \rho^{e'}(y) \\
& + \tfrac{i}{2}\, f^{ad'd}\, f^{c'eg}\, U^{d'e'}(y)\, U^{ac'}(w')\, U^{de}(x')\, \rho^g(x')\, \rho^{e'}(y)\Big] \\
+ (-p_8)\mathcal{K}_{\rm IV}\big|_{y\to x}\Big[& - \tfrac{i}{2}\, f^{cbd}\, f^{dd'g}\, U^{ab'}(x')\, U^{c'd'}(x)\, U^{c'b}(z)\, U^{ac}(w)\, \rho^{b'}(x')\, \rho^g(x) \\
& - \tfrac{i}{2}\, f^{acd}\, f^{d'eg}\, U^{ab'}(x')\, U^{c'd'}(x)\, U^{c'b}(z)\, U^{bc}(z)\, U^{de}(x)\, \rho^{b'}(x')\, \rho^g(x) \\
& + \tfrac{i}{2}\, f^{cbd}\, f^{de'g}\, U^{bd'}(z)\, U^{d'e'}(x)\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{c'}(x')\, \rho^g(x) \\
& - \tfrac{i}{2}\, f^{ad'd}\, f^{ee'g}\, U^{d'e'}(x)\, U^{ac'}(w')\, U^{de}(x)\, \rho^{c'}(x')\, \rho^g(x)\Big]
\end{aligned} \tag{9.7}$$

Group III, Row II, colour $\mathcal{O}_{\rm G3II}$ (+c.c.).

$$\begin{aligned}
\mathcal{O}^{(2)}\text{-part} = (-p_8)\mathcal{K}_{\rm IV}\big|_{y\to x'}\Big[& - \tfrac{i}{2}\, f^{cbd}\, f^{b'd'g}\, U^{c'd'}(x')\, U^{ab'}(x')\, U^{bc'}(z)\, U^{ac}(w)\, \rho^g(x')\, \rho^d(x) \\
& + \tfrac{i}{2}\, f^{ac'd}\, f^{b'd'g}\, U^{c'd'}(x')\, U^{ab'}(x')\, U^{de}(x)\, \rho^g(x')\, \rho^e(x) \\
& - \tfrac{i}{2}\, f^{cbd}\, f^{bb'g}\, U^{ab'}(x')\, U^{ac}(w)\, \rho^g(x')\, \rho^d(x) \\
& + \tfrac{i}{2}\, f^{acd}\, f^{bb'g}\, U^{ab'}(x')\, U^{bc}(z)\, U^{de}(x)\, \rho^g(x')\, \rho^e(x)\Big] \\
+ (-p_8)\mathcal{K}_{\rm IV}\big|_{y\to x}\Big[& - \tfrac{i}{2}\, f^{cbd}\, f^{dd'g}\, U^{c'd'}(x)\, U^{ab'}(x')\, U^{bc'}(z)\, U^{ac}(w)\, \rho^{b'}(x')\, \rho^g(x) \\
& - \tfrac{i}{2}\, f^{ac'd}\, f^{d'eg}\, U^{c'd'}(x)\, U^{ab'}(x')\, U^{de}(x)\, \rho^{b'}(x')\, \rho^g(x)
\end{aligned}$$

$$
\begin{aligned}
&+\tfrac{i}{2}\,f^{acd}\,f^{beg}\,U^{ab'}(x')\,U^{bc}(z)\,U^{de}(x)\,\rho^{b'}(x')\,\rho^{g}(x)\\
&-\tfrac{i}{2}\,N_c\;U^{ab'}(x')\,U^{ac}(w)\,\rho^{b'}(x')\,\rho^{c}(x)\Big]\\
&+(-p_8)\mathcal{K}_{\rm IV}\big|_{x\to x'}\Big[\tfrac{i}{2}\,f^{cbd}\,f^{b'dg}\,U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,U^{ac}(w)\,\rho^{g}(x')\,\rho^{d'}(y)\\
&-\tfrac{i}{2}\,f^{ac'd}\,f^{b'eg}\,U^{c'd'}(y)\,U^{ab'}(x')\,U^{de}(x')\,\rho^{g}(x')\,\rho^{d'}(y)\\
&-\tfrac{i}{2}\,f^{cbd}\,f^{b'dg}\,U^{ab'}(x')\,U^{ac}(w)\,\rho^{g}(x')\,\rho^{b}(y)\\
&+\tfrac{i}{2}\,f^{acd}\,f^{b'eg}\,U^{ab'}(x')\,U^{bc}(z)\,U^{de}(x')\,\rho^{g}(x')\,\rho^{b}(y)\Big]
\end{aligned}
\tag{9.8}
$$

Group III, Row III part 1, colour $\mathcal{O}_{\rm III(1)}$ (+c.c.).

$$
\begin{aligned}
\mathcal{O}^{(2)}\text{-part} = \tfrac{c_{\rm III(1)}}{k^+}\mathcal{K}_C\big|_{y\to x'}\Big[&-i\,f^{c'd'a}\,f^{ce'g}\,U^{e'c'}(y')\,U^{d'b}(z)\,U^{ac}(w)\,\rho^{g}(x')\,\rho^{b}(x)\\
&+i\,f^{c'd'a}\,f^{de'g}\,U^{e'c'}(y')\,U^{d'b}(z)\,U^{bc}(z)\,U^{ad}(x')\,U^{ce}(x)\,\rho^{g}(x')\,\rho^{e}(x)\\
&+i\,f^{c'd'a}\,f^{ce'g}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{ac}(w)\,\rho^{g}(x')\,\rho^{b}(x)\\
&-i\,f^{c'd'a}\,f^{de'g}\,U^{c'e'}(x')\,U^{ad}(x')\,U^{d'e}(x)\,\rho^{g}(x')\,\rho^{e}(x)\Big]\\
+\tfrac{c_{\rm III(1)}}{k^+}\mathcal{K}_C\big|_{x\to x'}\Big[&-i\,f^{c'd'a}\,f^{be'g}\,U^{e'c'}(y')\,U^{d'b}(z)\,U^{ac}(w)\,\rho^{g}(x')\,\rho^{c}(y)\\
&+i\,f^{c'd'a}\,f^{ee'g}\,U^{e'c'}(y')\,U^{d'b}(z)\,U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x')\,\rho^{g}(x')\,\rho^{d}(y)\\
&+i\,f^{c'd'a}\,f^{be'g}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{ac}(w)\,\rho^{g}(x')\,\rho^{c}(y)\\
&-i\,f^{c'd'a}\,f^{ee'g}\,U^{c'e'}(x')\,U^{ad}(y)\,U^{d'e}(x')\,\rho^{g}(x')\,\rho^{d}(y)\Big]
\end{aligned}
\tag{9.9}
$$

Group III, Row V, colour $\mathcal{O}_{\rm V}$ of Eq. (7.11) (+c.c.).

$$
\begin{aligned}
\mathcal{O}^{(2)}\text{-part} = \tfrac{c_{\rm V}}{k^+}\mathcal{K}_C\big|_{x\to x'}\Big[&-\tfrac{i}{2}\,f^{c'ca}\,f^{ee'g}\,U^{e'c'}(y')\,U^{ce}(x')\,U^{ad}(y)\,\rho^{g}(x')\,\rho^{d}(y)\\
&+\tfrac{i}{2}\,f^{c'ca}\,f^{ee'g}\,U^{e'c'}(y')\,U^{ad}(w)\,U^{ce}(x')\,\rho^{g}(x')\,\rho^{d}(y)\\
&+\tfrac{i}{2}\,f^{c'ca}\,f^{ee'g}\,U^{c'e'}(x')\,U^{ce}(x')\,U^{ad}(y)\,\rho^{g}(x')\,\rho^{d}(y)\\
&-\tfrac{i}{2}\,f^{c'ca}\,f^{ee'g}\,U^{c'e'}(x')\,U^{ad}(w)\,U^{ce}(x')\,\rho^{g}(x')\,\rho^{d}(y)\Big]\\
+\tfrac{c_{\rm V}}{k^+}\mathcal{K}_C\big|_{y\to x'}\Big[&-\tfrac{i}{2}\,f^{c'ca}\,f^{de'g}\,U^{e'c'}(y')\,U^{ce}(x)\,U^{ad}(x')\,\rho^{g}(x')\,\rho^{e}(x)\\
&+\tfrac{i}{2}\,f^{c'ca}\,f^{de'g}\,U^{e'c'}(y')\,U^{ad}(w)\,U^{ce}(x)\,\rho^{g}(x')\,\rho^{e}(x)\\
&+\tfrac{i}{2}\,f^{c'ca}\,f^{de'g}\,U^{c'e'}(x')\,U^{ce}(x)\,U^{ad}(x')\,\rho^{g}(x')\,\rho^{e}(x)\\
&-\tfrac{i}{2}\,f^{c'ca}\,f^{de'g}\,U^{c'e'}(x')\,U^{ad}(w)\,U^{ce}(x)\,\rho^{g}(x')\,\rho^{e}(x)\Big]\\
+\tfrac{c_{\rm V}}{k^+}\mathcal{K}_C\big|_{y\to x}\Big[&-\tfrac{i}{2}\,f^{c'ca}\,f^{deg}\,U^{e'c'}(y')\,U^{ce}(x)\,U^{ad}(x)\,\rho^{e'}(x')\,\rho^{g}(x)\\
&+\tfrac{i}{2}\,f^{c'ca}\,f^{deg}\,U^{e'c'}(y')\,U^{ad}(w)\,U^{ce}(x)\,\rho^{e'}(x')\,\rho^{g}(x)\\
&+\tfrac{i}{2}\,f^{c'ca}\,f^{deg}\,U^{c'e'}(x')\,U^{ce}(x)\,U^{ad}(x)\,\rho^{e'}(x')\,\rho^{g}(x)\\
&-\tfrac{i}{2}\,f^{c'ca}\,f^{deg}\,U^{c'e'}(x')\,U^{ad}(w)\,U^{ce}(x)\,\rho^{e'}(x')\,\rho^{g}(x)\Big]
\end{aligned}
\tag{9.10}
$$

Group III, Row VI, colour $\mathcal{O}_{\rm VI}$ (+c.c.).

$$
\begin{aligned}
\mathcal{O}^{(2)}\text{-part} = \tfrac{c_{\rm VI}}{k^+}\mathcal{K}_C\big|_{y\to x'}\Big[&-\tfrac{i}{2}\,f^{c'd'a}\,f^{de'g}\,U^{e'c'}(y')\,U^{d'b}(z)\,U^{bc}(z)\,U^{ad}(x')\,U^{ce}(x)\,\rho^{g}(x')\,\rho^{e}(x)\\
&+\tfrac{i}{2}\,f^{c'd'a}\,f^{de'g}\,U^{e'c'}(y')\,U^{d'b}(z)\,U^{ad}(x')\,\rho^{g}(x')\,\rho^{b}(x)\\
&+\tfrac{i}{2}\,f^{c'd'a}\,f^{de'g}\,U^{c'e'}(x')\,U^{ad}(x')\,U^{d'e}(x)\,\rho^{g}(x')\,\rho^{e}(x)\\
&-\tfrac{i}{2}\,f^{c'd'a}\,f^{de'g}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{ad}(x')\,\rho^{g}(x')\,\rho^{b}(x)\Big]\\
+\tfrac{c_{\rm VI}}{k^+}\mathcal{K}_C\big|_{x\to x'}\Big[&-\tfrac{i}{2}\,f^{c'd'a}\,f^{ee'g}\,U^{e'c'}(y')\,U^{d'b}(z)\,U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x')\,\rho^{g}(x')\,\rho^{d}(y)\\
&+\tfrac{i}{2}\,f^{c'd'a}\,f^{be'g}\,U^{e'c'}(y')\,U^{d'b}(z)\,U^{ad}(y)\,\rho^{g}(x')\,\rho^{d}(y)
\end{aligned}
$$

$$
\begin{aligned}
&+ \tfrac{i}{2}\, f^{c'd'a}\, f^{ee'g}\, U^{c'e'}(x')\, U^{ad}(y)\, U^{d'e}(x')\, \rho^g(x')\, \rho^d(y) \\
&- \tfrac{i}{2}\, f^{c'd'a}\, f^{be'g}\, U^{bd'}(z)\, U^{c'e'}(x')\, U^{ad}(y)\, \rho^g(x')\, \rho^d(y)\Big] \\
&+ \tfrac{c_{\rm VI}}{k^+}\mathcal{K}_C\big|_{y\to x}\Big[\tfrac{i}{2}\, f^{c'd'a}\, f^{deg}\, U^{e'c'}(y')\, U^{d'b}(z)\, U^{bc}(z)\, U^{ad}(x)\, U^{ce}(x)\, \rho^{e'}(x')\, \rho^g(x) \\
&+ \tfrac{i}{2}\, f^{c'd'a}\, f^{bdg}\, U^{e'c'}(y')\, U^{d'b}(z)\, U^{ad}(x)\, \rho^{e'}(x')\, \rho^g(x) \\
&- \tfrac{i}{2}\, f^{c'd'a}\, f^{deg}\, U^{c'e'}(x')\, U^{ad}(x)\, U^{d'e}(x)\, \rho^{e'}(x')\, \rho^g(x) \\
&- \tfrac{i}{2}\, f^{c'd'a}\, f^{bdg}\, U^{bd'}(z)\, U^{c'e'}(x')\, U^{ad}(x)\, \rho^{e'}(x')\, \rho^g(x)\Big]
\end{aligned}
\tag{9.11}
$$

10 Four ρ's (unchanged; Q2)

The four-ρ p^+ integrands are exactly $\propto \frac{1}{p^+}$.

- Group I Rows I and III: range $(\Lambda, \vee)$.
- Group I Row IV, Group II Rows I, II, III, Group III Rows I, II, IV: range $(\Lambda, \vee - k^+)$, which gives $\log\frac{\vee}{\Lambda} + \log(1-a)$.

The $\log\frac{\vee}{\Lambda}$ parts cancel after symmetrization. The $\log(1-a)$ parts do not (max. $|\Sigma| = 12$–24 at random points). Exactly, the four-ρ finite term is

$$
\frac{d^3N_{4\rho}}{d^3k} = \mathcal{N}\log\Big(1-\frac{k^+}{\vee}\Big) \times \Big[\text{the lines of Eq. (4rho) of } \texttt{4rho.tex} \text{ with range } \vee - k^+\Big] \tag{10.1}
$$

(G1 IV, G2 I, II, III, G3 I, II, IV; c.c. of the Group I and III lines; written out in `Finite_terms_exact.pdf`, Eq. (10.1)). It is zero if the independently emitted gluon has $p^+ < \vee$ in every row (Q2).

11 Final result

For hadron production ($\overline{MS}$, $d_\perp = 2-\epsilon$, $n_f = 0$; $a = \frac{k^+}{\vee}$, $L_* = \log\frac{\sqrt{\vee\Lambda}}{k^+}$):

$$
\begin{aligned}
\frac{d^3N_h}{d^3k} &= \int_{k^+/\vee}^1 \frac{d\zeta}{\zeta^2}\, D_{h/g}(\zeta,\mu_F)\, \frac{d^3N_g}{d^3k}\Big(\frac{k^+}{\zeta}, \frac{\mathbf{k}}{\zeta}\Big), \\
\frac{d^3N_g}{d^3k} &= \mathrm{LO}\big|_{\alpha_s(\mu_F)} + \mathcal{N}\frac{1}{2}\log\frac{\vee}{\Lambda}\,(\mathcal{J}+\mathcal{B}) \qquad \text{(evolution, } \texttt{Evolution_terms_corrected.tex}\text{)} \\
&\quad + \frac{\alpha_s}{2\pi}N_c\Big[\frac{67}{9} - \frac{\pi^2}{3}\Big]\mathrm{LO} \qquad \text{(running coupling, (4.3))} \\
&\quad - \frac{\alpha_s}{2\pi}N_c \log(1-a)\log\frac{1-a}{a}\,\mathrm{LO} \qquad (V_\vee,\ (4.3);\ \text{with Q1}) \\
&\quad + \frac{\alpha_s N_c}{2\pi}\, L_*\,\mathrm{LO}\Big[\log\frac{(y-y')^4}{(x-y)^2(x'-y')^2}\Big] - \frac{\alpha_s}{2\pi}\int_a^1 \frac{d\xi}{\xi^2} P^{(r)}_{gg,+}(\xi)\,\mathrm{LO}_\xi\big[\ell_\Delta(\mu_F)\big] + \frac{d^3N_{\rm lin}}{d^3k} \\
&\quad + \frac{\alpha_s N_c}{4\pi}\log(1-a)\,\mathrm{LO}\big[\hat{E}_X + \hat{E}_{X'}\big] \qquad \text{(Q1)} \\
&\quad + \Big[(7.6)+(7.7)+(7.9)+(7.12)\Big]_{\text{two }\rho} \\
&\quad + \Big[(7.1)+(7.2)+(7.4)+(8.1)+(8.3)\Big]_{\text{three }\rho = S + \text{two-}\rho\text{ parts, Sec. 9}} \\
&\quad + \frac{d^3N_{4\rho}}{d^3k} \qquad \text{(Q2)} \qquad + O(\alpha_s^3) + O\Big(\frac{\Lambda}{k^+}\Big).
\end{aligned}
\tag{11.1}
$$

- Group I and Group III terms include their c.c.; Group II terms (in (7.9), (8.1) through block B, the DGLAP terms) have none.
- Every L_* term together is $\mathcal{N}\log\frac{k^+}{\sqrt{\vee\Lambda}}(\mathcal{J}-\mathcal{B})$, Eq. (6.2). All of them vanish at mid rapidity, $k^+ = \sqrt{\vee\Lambda}$.
- Nothing is expanded in $\frac{k^+}{\vee}$. Λ appears only in $\log\frac{\vee}{\Lambda}$ and in L_*.
- LO is $O(\alpha_s)$; every other term is $O(\alpha_s^2)$.
- With your earlier convention (evolution $\mathcal{N}\log\frac{\vee}{\Lambda}\mathcal{J}$): $L_* \to \log\frac{\vee}{k^+}$ everywhere.

12 Open points

Q1 The pole of Eq. (5.3). $\frac{\alpha_s}{2\pi}\frac{2}{\epsilon}N_c\log(1-\frac{k^+}{\vee})$LO is left over.

- It comes from $\log(\frac{\vee}{k^+}-1)$ in Group I Row II part 1, i.e. the limit $\xi \geq \frac{k^+}{\vee}$ of the dressed one-ρ state.
- Is that limit intended for this virtual term (in AABKL it is $\int_0^1 d\xi$), or is a real term with the same limit missing?

Q2 Four ρ's, Eq. (10.1): is the range of the independently emitted gluon $(\Lambda, \vee - k^+)$ or $(\Lambda, \vee)$?

Q3–Q11 The other questions of `Finite_terms_exact.pdf` (Q3–Q11) are unchanged, except Q5: the two-ρ parts of the three-ρ rows are now written out (Sec. 9).

13 Checks

Scripts in `nlo_gluon_paper/finite_v2/` (they use the transcription and closed forms of `../finite/`):

- `split.py`: Sec. 6 and the c.c. test;
- `decomp.py`, `compact.py`: Sec. 9;
- `final_checks.py`: everything else;
- `make_note.py`: this note.

check	result
closed forms of Sec. 7 ($\Lambda = 0$) vs. your file ($\Lambda = 10^{-10}$), $\vee/k^+ = 1.3, 2.5, 10, 300$	$\leq 2.6 \times 10^{-9} = O(\Lambda/k^+)$
master integrals $\mathcal{M}_{0,1,2}$ vs. quadrature	3.0×10^{-11}
Group II Row I part 2 collected, Eq. (7.9), vs. 2a–2f	1.2×10^{-15}
split: $R_{\rm row}(a)$ constant for $a = 10^{-4} \ldots 10^{-8}$	spread $\leq 8.2 \times 10^{-4}$
c.c.: Group II self-conjugate / Group I, III not	1.7×10^{-15} / ≥ 1
symmetric colour forms, Sec. 9 and Eqs. (8.2), (8.4)	$\leq 2.9 \times 10^{-14}$
two-ρ parts vs. the reordering engine	$\leq 1.9 \times 10^{-15}$
poles, $\delta(1-\xi)$ coefficient (`sympy`)	$11/6 - \log(1-a)$
β_0 in Rows II + V; finite $\hat{E}$ terms	0; 0
coefficient of $L_\vee$ in Rows II, V(UV), G2 IV	$\frac{\alpha_s N_c}{2\pi}(2\ell_\Delta - \ell_X - \ell_{X'})$